

Practice Problem 9.4

Express these sinusoids as phasors:

时域信号→相量信号:

统一表示成cos形式, 再用幅度和相位构成相量(复数)

(a) $v = -14 \sin(5t - 22^\circ)$ V

(b) $i = -8 \cos(16t + 15^\circ)$ A

Answer: (a) $V = 14 \angle 68^\circ$ V, (b) $I = 8 \angle -165^\circ$ A.

$$\therefore -\sin \theta = \cos(\theta + 90^\circ)$$

$$-\cos \theta = \cos(\theta + 180^\circ)$$

$$\therefore -14 \sin(5t - 22^\circ) = 14 \cos(5t + 68^\circ) \Rightarrow V = 14 \angle 68^\circ \text{ V}$$

$$-8 \cos(16t + 15^\circ) = 8 \cos(16t + 195^\circ) \Rightarrow I = 8 \angle 195^\circ \text{ A}$$

$$\text{或 } 8 \angle -165^\circ \text{ A}$$

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Find the sinusoids corresponding to these phasors:

(a) $V = -25 \angle 40^\circ$ V

(b) $I = j(12 - j5)$ A

Answer: (a) $v(t) = 25 \cos(\omega t - 140^\circ)$ V or $25 \cos(\omega t + 220^\circ)$ V,

(b) $i(t) = 13 \cos(\omega t + 67.38^\circ)$ A.

Practice Problem 9.5

相量信号→时域信号:

• 先将相量表示成极坐标形式, 再写成cos形式:

• 或 $v(t) = \operatorname{Re}(V e^{j\omega t})$

$$\textcircled{1} v(t) = -25 \cos(\omega t + 40^\circ)$$

$$= 25 \cos(\omega t - 140^\circ) \text{ 或 } 25 \cos(\omega t + 220^\circ)$$

$$\textcircled{2} 5 + j12 = 13 \angle 67.38^\circ$$

$$\Rightarrow i(t) = 13 \cos(\omega t + 67.38^\circ) \text{ A}$$

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Practice Problem 10.1Using nodal analysis, find v_1 and v_2 in the circuit of Fig. 10.3.

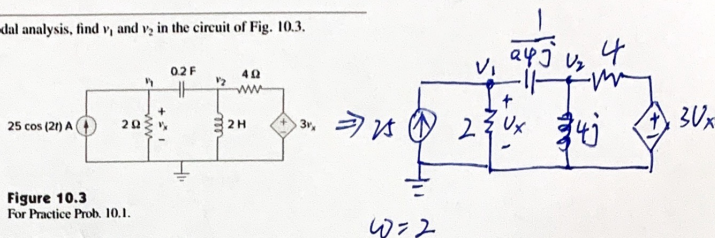
电路的频域分析法:

- ① 时域信号 $\rightarrow$ 频域信号;
- ② 电路元件用阻抗表达, 在频域中分析电路;
- ③ 频域信号 $\rightarrow$ 时域信号

频域节点电压法

Figure 10.3
For Practice Prob. 10.1.

Answer: $v_1(t) = 28.31 \cos(2t + 60.01^\circ)$ V,
 $v_2(t) = 82.56 \cos(2t + 57.12^\circ)$ V.



$$25 = \frac{V_1}{2} + (V_1 - V_2) \cdot 0.4j$$

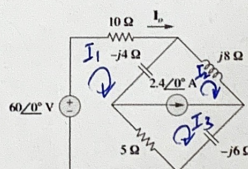
$$(V_1 - V_2) \cdot 0.4j = \frac{V_2}{4j} + \frac{V_2 - 3V_1}{4}$$

$$\Rightarrow \begin{bmatrix} 0.5 + 0.4j & -0.4j \\ 0.75 + 0.4j & -0.25 - 0.15j \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 25 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 28.318 \angle 60.018^\circ \\ 82.559 \angle 57.127^\circ \end{bmatrix} \Rightarrow$$

$$V_1(t) = 28.318 (\cos 2t + 60.018^\circ) \text{ V}$$

$$V_2(t) = 82.559 (\cos 2t + 57.127^\circ) \text{ V}$$

Calculate current I_o in the circuit of Fig. 10.11.Answer: $6.089/5.94^\circ$ A.**Practice Problem 10.4****Figure 10.11**
For Practice Prob. 10.4.

频域网孔电流法~supermesh

$$-60 + (15 - 4j)I_1 - (-4j)I_2 - 5I_3 = 0$$

$$4jI_2 + (5 - 6j)I_3 + 4jI_1 - 5I_1 = 0$$

$$I_3 - I_2 = 2.4$$

$$\Rightarrow \begin{bmatrix} 15 - 4j & 4j & -5 \\ -5 + 4j & 4j & 5 - 6j \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 2.4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 6.09 \angle 5.943^\circ \\ 4.058 \angle 4.015^\circ \\ -6.454 \angle 2.523^\circ \end{bmatrix} \Rightarrow I_0 = I_1 = 6.09 \angle 5.943^\circ \text{ A}$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.

频域叠加定理:

注意, 电路中若有不同的频率, 只能时域叠加, 不能相量叠加

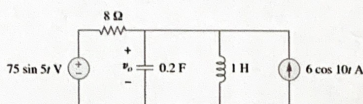
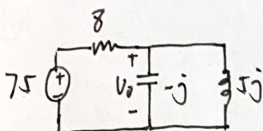


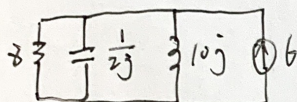
Figure 10.15
For Practice Prob. 10.6.

Answer: $11.577 \sin(5t - 81.12^\circ) + 3.154 \cos(10t - 86.24^\circ) \text{ V}$.

① $\omega = 5$



② $\omega = 10$



$$\text{对 } \textcircled{1} \quad v_o = \frac{75}{8 + \frac{5}{4j}} \cdot \frac{5}{4j} = 11.578 \angle -81.119^\circ$$

$$\Rightarrow v_o(t) = 11.578 \sin(5t - 81.119^\circ) \text{ V}$$

$$\text{对 } \textcircled{2} \quad v_o = 6 \cdot \frac{1}{\frac{1}{8} + 2j + \frac{1}{10j}} = 3.151 \angle -86.236^\circ$$

$$\Rightarrow v_o(t) = 3.151 \cos(10t - 86.236^\circ) \text{ V}$$

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$$\therefore v_o(t) = 11.578 \sin(5t - 81.119^\circ) + 3.154 \cos(10t - 86.236^\circ) \text{ V}$$

Determine the Thevenin equivalent of the circuit in Fig. 10.27 as seen from the terminals a-b.

Answer: $Z_{Th} = 4.473 \angle -7.64^\circ \Omega$, $V_{Th} = 11.763 \angle 72.9^\circ \text{ volts}$.

Practice Problem 10.9

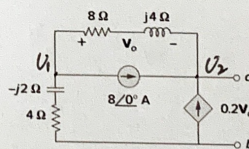


Figure 10.27
For Practice Prob. 10.9.

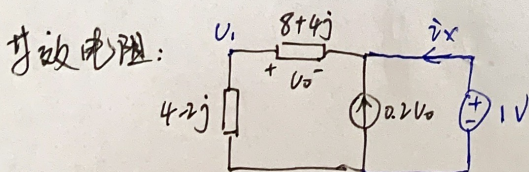
频域戴维南定理

开路电压: $\frac{V_1}{4-j2} + 8 + \frac{V_1-V_2}{8+4j} = 0$

$$8 + 0.2(V_1 - V_2) + \frac{V_1 - V_2}{8 + 4j} = 0$$

$$\Rightarrow \begin{bmatrix} \frac{1}{4-j2} + \frac{1}{8+4j} & \frac{-1}{8+4j} \\ 0.2 + \frac{1}{8+4j} & -0.2 - \frac{1}{8+4j} \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -8 \\ -8 \end{bmatrix} \Rightarrow V_{Th} = V_2 = 11.763 \angle 72.89^\circ$$

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$$\begin{cases} \frac{V_1 - 1}{8 + 4j} + 0.2(V_1 - 1) + i_x = 0 \\ \frac{V_1}{4 - j2} + \frac{V_1 - 1}{8 + 4j} = 0 \end{cases}$$

$$\Rightarrow Z_{Th} = \frac{1}{i_x} = 4.472 \angle -7.64^\circ \Omega$$